Qingkai Dong

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Education

University of Connecticut Aug 2023 – Present
PhD in Statistics Storrs, CT, USA
Advisors: Prof. HaiYing Wang and Prof. Jun Yan Passed the Ph.D. Qualifying Exam

Zhongnan University of Economics and Law Sep 2020 – May 2023
MS in Mathematical Statistics Wuhan, China
Thesis: Model Averaging and Variable Selection for Accelerated Failure Time Models

Qingdao University Sep 2016 – May 2020
BS in Applied Statistics Qingdao, China

Research Experience

Research Assistant (Statistical Methodology) Aug 2023 – Present
University of Connecticut

- **Primary output:** First-authored methodology paper on rare-feature-aware subsampling for regression models; manuscript in preparation for journal submission.
- Developed theory showing that estimation efficiency for rare binary covariate coefficients is driven by the limited number of rare observations, explaining numerical instability and slow convergence.
- Introduced a balanced subsampling method that quantifies rarity to ensure adequate representation of rare features without pilot sampling, improving robustness and efficiency.
- Implemented the methods in the subsampling R package; validated performance via theoretical results, simulation studies, and real-data applications; produced documentation and reproducible examples.

Research Assistant (Academia–Industry Collaboration) Mar 2024 – Dec 2025
Servier Pharmaceuticals & University of Connecticut

- **Primary output:** Coauthored a manuscript introducing a predictive-modeling–assisted interim analysis framework for censored time-to-event endpoints; manuscript in preparation for journal submission.
- Developed a covariate-informed approach that augments interim data by predicting event times for censored participants, improving estimation of treatment effects and conditional power.
- Proposed evaluation metrics for conditional power forecast accuracy and futility decision quality; conducted extensive simulations to characterize when performance improves or degrades under varied censoring, covariate informativeness/type, prediction range, and sample size.

Research Assistant (Real-World Health Data) Jul 2024 – Aug 2024
University of Connecticut

- Conducted data preprocessing and statistical analyses on All of Us data to study associations among social determinants of health, frailty index, and accelerated aging in breast cancer patients.
- Delivered analysis-ready datasets and reproducible scripts; performed exploratory and confirmatory analyses under missingness and data-quality constraints.

Research Project (Chemical Sensor Data Analytics) Mar 2024 – May 2024
University of Connecticut

- Analyzed fluorescence response data from porphyrinoid sensors; applied clustering and statistical analysis to improve compound differentiation and sensor selection.
- Produced interpretable summaries and visualizations to support experimental decision-making.

Cross-Disciplinary Collaboration (LLM Explainability) 2025 – Present
University of Connecticut & collaborators

- Coauthored a survey manuscript on time series explainability with an emphasis on LLM-enabled semantic explanations; curated benchmarks and an accompanying repository.

Software & Open-Source

R package subsampling (CRAN): A statistical computing toolkit for optimal subsampling in big-data modeling. Supports GLMs, quantile regression, and rare event/rare feature regimes.

- Implemented optimized sampling probability computation targeting variance reduction and prediction accuracy, including practical routines for pilot sampling and two-step designs.

- Engineering focus: modular API, documentation/vignettes, reproducible examples, and scalable computation patterns.

Teaching Experience

Instructor Aug 2025 – May 2026
University of Connecticut

- **STAT 3675Q:** Statistical Computing — comprehensive statistical computing in R/RStudio, including data acquisition and management, visualization, reproducible workflows, and core analyses in regression, nonparametric inference, GLMs, categorical data analysis, time series, and classification.

Academic Presentations

Invited Talk - New England Statistics Symposium (NESS) 2026
Poster Presentation - New England Statistics Symposium (NESS) 2026
Poster Presentation - The Design and Analysis of Experiments (DAE) 2026
Poster Presentation - New England Rare Disease Statistics (NERDS) Workshop, Boston, MA Oct 2025
Poster Presentation - Dahshu Data Science Symposium, Storrs, CT Oct 2025

Academic Service

Reviewer: *Computational Statistics and Data Analysis*; *Statistical Papers*; *Sankhya B*; *Journal of Systems Science and Mathematical Sciences (Chinese)*

Awards

New England Statistics Symposium (NESS) Poster Award 2026
ICSA Applied Statistics Symposium Student Paper Award 2026
Predoc Fellowship at University of Connecticut 2025
First-class Scholarship at Zhongnan University of Economics and Law 2020, 2022

Technical Skills

Statistics/Methods: survival analysis; interim analysis & conditional power; design simulation; big-data subsampling; model averaging; variable selection

Programming: R (package development), Python (machine learning / deep learning), Git, LaTeX

Selected Publications

Dong Q, Liu B, Zhao H. Weighted Least Squares Model Averaging for Accelerated Failure Time Models. *Computational Statistics and Data Analysis*, 2023.

Zhao H, Dong Q. Variable Selection for Additive Hazards Model with Current Status Data. *Journal of Systems Science and Mathematical Sciences*, 2022.

Zhao H, Liu B, Dong Q. Jackknife Model Averaging of AFT Model with Current Status Data. *Acta Mathematicae Applicatae Sinica*, 2023.

Chen Z, Lucchesi G, Dong Q, Zheng X, Song D, Wen Q, Cheng W, Ni J, Luo D. From Signals to Semantics: A Survey on Time Series Explainability through a Human-Cognitive Lens. *Under review*, 2025.